**MTH634 – Topology (Quin No.4)**

**Question # 2 of 10 ( Start time: 12:53:08 PM, 03 July 2025 )**

If $(X, \tau)$ be a separable topology then it must have countable dense set.

**Select the correct option**

| | |
|---|---|
| ◉ | True |
| ○ | False |

**MTH634 - Topology (Quin No.4)**

**Question # 6 of 10 ( Start time: 12:47:53 PM, 03 July 2025 )**

Let X be a topological space. Which of the following statement is true?

**Select the correct option**

| | |
|---|---|
| ○ | Every pair of disjoint closed sets in X can be separated by closed sets. |
| ◉ | Every pair of disjoint closed sets in X can be separated by open sets. |

**MTH634 - Topology (Quin No.4)**

**Question # 9 of 10 ( Start time: 12:55:24 PM, 03 July 2025 )**

A topological space is a $T_1$ - space if and only if each of its finite subset is ....

**Select the correct option**

| | |
|---|---|
| ◉ | a closed set. |
| ○ | both open and closed. |
| ○ | an open set |
| ○ | neither open nor closed. |

**MTH634 - Topology (Quin No.4)**

**Question # 8 of 10 ( Start time: 12:55:09 PM, 03 July 2025 )**

Which of the following statements is true about metric spaces?

**Select the correct option**

| | |
|---|---|
| ◉ | Every metric space is Hausdorff. |
| ○ | Every metric space is complete. |
| ○ | Every metric space is not Hausdorff. |
| ○ | Every metric space is discrete. |

**MTH634 - Topology (Quin No.4)**

**Question # 3 of 10 ( Start time: 12:46:57 PM, 03 July 2025 )**

A topological space is a $T_1$ - space if and only if each of its finite subset is ....

**Select the correct option**

| | |
|---|---|
| ○ | neither open nor closed. |
| ○ | both open and closed. |
| ○ | an open set |
| ● | a closed set. |

**MTH634 - Topology (Quin No.4)**

**Question # 5 of 10 ( Start time: 12:54:18 PM, 03 July 2025 )**

Which of the following space is a Housdorff?

**Select the correct option**

| | |
|---|---|
| ◉ | Real numbers with Lower limit topology. |
| ○ | A non empty set X with indiscrete space. |
| ○ | Real numbers with toplogy generated by $\{(a,\infty)\|a \in R\}$ |
| ○ | An Infinite set with cofinite toplogy. |

**MTH634 - Topology (Quin No.4)**

**Question # 10 of 10 ( Start time: 12:49:09 PM, 03 July 2025 )**

Which of the following space is a Housdorff?

**Select the correct option**

| | |
|---|---|
| ○ | An infinite set with cofinite toplogy. |
| ● | Real numbers with Lower limit topology. |
| ○ | Real numbers with toplogy generated by $\{(a, \infty) \mid a \in R\}$ |
| ○ | A non empty set X with indiscrete space. |

**MTH634 - Topology (Quin No.4)**

**Question # 2 of 10 ( Start time: 12:46:44 PM, 03 July 2025 )**

Which of the following statements is true about metric spaces?

**Select the correct option**

| | |
|---|---|
| ○ | Every metric space is not Hausdorff. |
| ● | Every metric space is Hausdorff. |
| ○ | Every metric space is discrete. |
| ○ | Every metric space is complete. |

**MTH634 - Topology (Quin No.4)**

**Question # 4 of 10 ( Start time: 12:53:45 PM, 03 July 2025 )**

Which of the following statements is true about Hausdorff spaces?

**Select the correct option**

| | |
|---|---|
| ○ | Every open set is dense. |
| ● | Every pair of distinct points can be separated by disjoint open sets. |
| ○ | Every sequence has a convergent subsequence. |
| ○ | Every pair of distinct points can be separated by overlapping open sets. |

**MTH634 - Topology (Quin No.4)**

**Question # 10 of 10 ( Start time: 12:55:37 PM, 03 July 2025 )**

Which of the following is not an example of compact spaces?

**Select the correct option**

| | |
|---|---|
| ◉ | An infinite set with discrete topology. |
| ○ | A finite set with any toplogy. |
| ○ | A set with topology containing finite number of elements. |
| ○ | A set with indiscrete toplogy. |

**MTH634 - Topology (Quin No.4)**

**Question # 9 of 10 ( Start time: 12:48:58 PM, 03 July 2025 )**

Which of the following statements correctly characterizes a $T_1$ topological space?

**Select the correct option**

| | |
|---|---|
| ○ | Every subset is closed. |
| ◉ | Every singleton subset is closed. |
| ○ | Every subset is open. |
| ○ | Every singleton subset is open. |

**MTH634 - Topology (Quin No.4)**

**Question # 8 of 10 ( Start time: 12:48:36 PM, 03 July 2025 )**

Which of the following topological spaces is Hausdorff?

**Select the correct option**

| | |
|---|---|
| ○ | The set of all finite sets with the discrete topology. |
| ● | The set of all real numbers with the usual topology. |
| ○ | The set of all infinite sets with the discrete topology. |
| ○ | The set of all points in the plane with the Euclidean topology. |

**MTH634 - Topology (Quin No.4)**

**Question # 7 of 10 ( Start time: 12:48:22 PM, 03 July 2025 )**

Which of the following statements is true about discrete topological space?

**Select the correct option**

| | |
|---|---|
| ○ | Every discrete topological space may or may not be regular. |
| ○ | None of these |
| ● | Every discrete topological space is regular. |
| ○ | Every discrete topological space is not regular. |

**MTH634 - Topology (Quin No.4)**

**Question # 1 of 10 ( Start time: 12:52:52 PM, 03 July 2025 )**

Consider a set $X$ with the indiscrete topology. What can be said about the singleton sets in $X$?

**Select the correct option**

| | |
|---|---|
| ○ | They are closed but not open. |
| ○ | They are open but not closed. |
| ○ | They are open and closed. |
| ● | They are neither open nor closed. |

**MTH634 - Topology (Quin No.4)**

**Question # 1 of 10 ( Start time: 12:46:17 PM, 03 July 2025 )**

Which of the following statements is equivalent to the statement that a topological space X is T1?

**Select the correct option**

| | |
|---|---|
| ○ | For any two distinct points x and y in X, there exist disjoint open sets U and V such that x ∈ U and y ∈ V. |
| ○ | For any two distinct points x and y in X, there exists an open set U such that x ∈ U and y ∉ U. |
| ● | For any two distinct points x and y in X, there exists an open set V such that y ∈ V and x ∉ V. |
| ○ | None |

**MTH634 - Topology (Quin No.4)**

**Question # 7 of 10 ( Start time: 12:54:51 PM, 03 July 2025 )**

Given a Hausdorff space $(X, T)$ and a subset $Y \subset X$, with the subspace topology $(Y, T_Y)$, which of the following statements is true?

**Select the correct option**

| | |
|---|---|
| ○ | Y must be connected. |
| ● | Y with the subspace topology $(Y, T_Y)$ is also Hausdorff. |
| ○ | Y must be compact |
| ○ | Y with the subspace topology $(Y, T_Y)$ is not Hausdorff. |

Click to

**MTH634 - Topology (Quin No.4)**

**Question # 3 of 10 ( Start time: 12:53:24 PM, 03 July 2025 )**

Which of the following is a correct reason why every subspace of a Hausdorff space is also Hausdorff?

**Select the correct option**

| | |
|---|---|
| ○ | Because every subspace of a topological space is always Hausdorff. |
| ● | Because the subspace inherits the separation properties from the parent space. |
| ○ | Because every subspace is closed in a Hausdorff space. |
| ○ | Because the subspace have different separation properties from the parent space. |

**MTH634 - Topology (Quin No.4)** **Quiz Start Time: 12:52 PM, 03 July 2025**

**Question # 6 of 10 ( Start time: 12:54:38 PM, 03 July 2025 )** **Total Marks:** 1

Which of the following statements is equivalent to the statement that a topological space X is regular?

**Select the correct option**

Reload Math Equations

| | |
|---|---|
| ● | A topological space X is said to be regular if, for every point x in X and every closed set F in X such that $x \notin F$, there exist disjoint open sets U and V in X such that $x \in U$ and $F \subseteq V$. |
| ○ | For any point x in X and any closed set F such that $x \notin F$, there exists a closed set G such that $x \in G$ and $F \cap G = \emptyset$. |

Click to Save Answer & Move to Next Question

MTH634 - Topology (Quin No.4) Quiz Start Time: 12:46 PM, 03 July 2025

Question # 5 of 10 ( Start time: 12:47:39 PM, 03 July 2025 ) Total Marks: 1

Which of the following statements is equivalent to the statement that a topological space X is regular?

Select the correct option

Reload Math Equations

- (●) A topological space X is said to be regular if, for every point x in X and every closed set F in X such that $x \notin F$, there exist disjoint open sets U and V in X such that $x \in U$ and $F \subseteq V$.
- ( ) For any point x in X and any closed set F such that $x \notin F$, there exists a closed set G such that $x \in G$ and $F \cap G = \emptyset$.

Click to Save Answer & Move to Next Question

MTH634 - Topology (Quin No.4) Quiz Start Time: 12:46 PM, 03 July 2025

Question # 4 of 10 ( Start time: 12:47:15 PM, 03 July 2025 ) Total Marks: 1

Which of the following statements is equivalent to the statement that a topological space X is regular?

Select the correct option

Reload Math Equations

- ( ) For any point x in X and any closed set F such that $x \notin F$, there exists a closed set G such that $x \in G$ and $F \cap G = \emptyset$.
- (●) A topological space X is said to be regular if, for every point x in X and every closed set F in X such that $x \notin F$, there exist disjoint open sets U and V in X such that $x \in U$ and $F \subseteq V$.

Click to Save Answer & Move to Next Question